

Interactive Visual Analytics of Carbon Cycle Science

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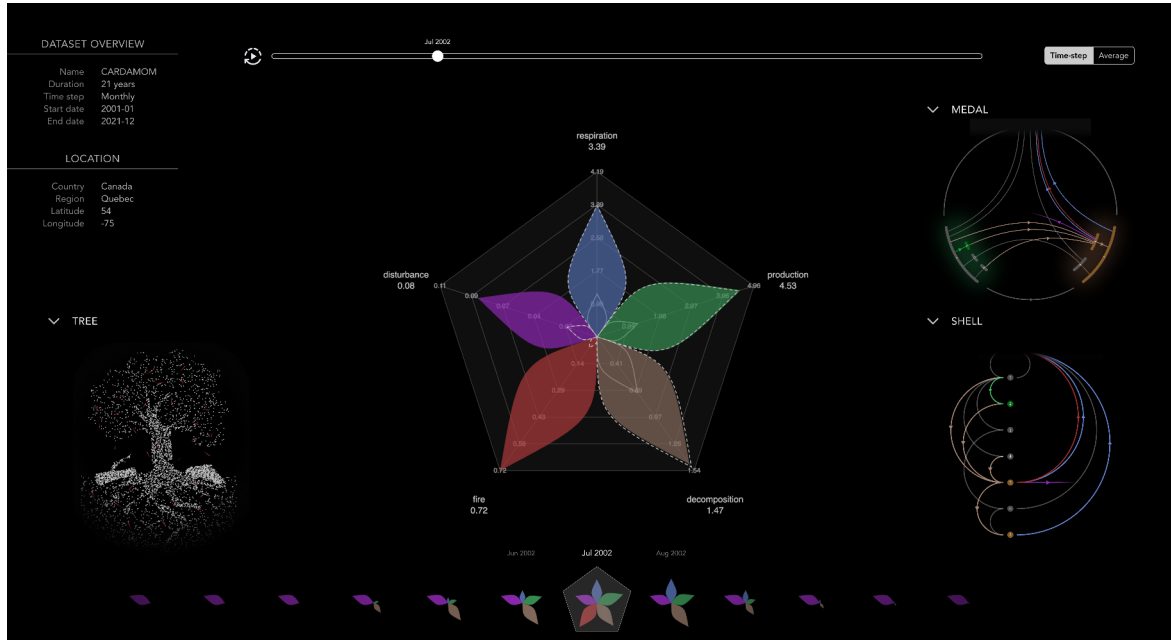


Figure 1: The CLOVE interface showing data from a major fire related event in June, 2002, and described in Section 5.

ABSTRACT

Tracking the flow of carbon (C) through the Earth’s terrestrial biosphere remains a major challenge to understanding how ecosystems respond to environmental change [11]. To build an understanding of this flow, researchers have recently developed models that include thousands of variables whose intricate inter-dependencies can make them difficult to interpret. To open this scientific black box, we partnered with C cycle scientists at the NASA Jet Propulsion Lab who developed the Carbon Data-Model (CARDAMOM) framework [5]. This paper presents a design study of CLOVE, a C science visual analytics application that encodes model dimensions in an intuitive way by using visual metaphors that correspond to the natural world. We describe how CLOVE encapsulates C states and fluxes in complex models, and explains how C cycle scientists can use it to diagnose their model outputs.

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1 INTRODUCTION

Recent advances in satellite Earth observations provide large datasets that describe Earth’s ecosystems at unprecedented scales and resolutions [29, 14, 19, 30], affording Earth scientists the opportunity to understand how carbon (C) cycles through the environment at unprecedented scale. Because C so strongly impacts climate, more accurate understanding of the C cycle is critical to assessments of the potential impacts of changes to the current C cycle equilibrium. However, integrating and interpreting these data is challenging due to complex interactions between variables, model uncertainties, and measurement errors.

To overcome these challenges, scientists have recently taken a Bayesian model-data fusion approach, enabling researchers to estimate C cycle states and processes with quantifiable uncertainties, independent of traditional plant functional type assumptions or steady-state constraints. This approach creates models that more accurately fit the data, but whose inherent complexity and high-dimensionality (over 100 interrelated variables) make them difficult to interpret, debug, and understand.

To support this scientific investigation, we partnered with scientists at the NASA Jet Propulsion Laboratory (NASA/JPL) working on the C Data-Model (CARDAMOM) Framework [5], a diagnostic modeling framework designed to integrate satellite observations

with terrestrial C cycle models. The overarching objective of CARDAMOM is to enhance scientific understanding of global C fluxes, C allocation patterns, residence times, and their environmental implications.

This paper presents CLOVE, a visual analytics approach to understanding C cycle data that represents relationships of C flux data using the CARDAMOM framework. CLOVE allows C cycle scientists to meaningfully interpret complex C cycle dynamics, facilitating exploration of key ecological relationships, patterns, and processes that would otherwise remain opaque or misunderstood. We present a design study [21] of CLOVE with the CARDAMOM modeling team at NASA/JPL that describes the users and their data and tasks, and then we unpack how CLOVE's visual encoding and interaction support those tasks. We argue that CLOVE is designed to enable researchers to rapidly grasp emergent patterns, relationships, and dynamics within global C datasets, ultimately supporting new insights and improved predictions of environmental change.

2 BACKGROUND

2.1 Carbon Cycle Science

One of the main challenges in C cycle science is to understand how C moves through the Earth system - between the atmosphere, vegetation, soils, and oceans - and how these flows are influenced by both natural processes and human activities [12]. The terrestrial C cycle remains a particularly uncertain component in the Earth system [3]. To reduce uncertainties, research efforts across scales strive to advance understanding of how C stores (e.g., foliage, woody biomass and soils) are modulated by observed fluxes (photosynthesis, respiration, and disturbance fluxes) to track how C is stored, transferred, and released over time [5, 19, 28, 16, 8, 15]. To understand the emergent behavior of terrestrial C cycling under nominal and hypothetical environmental scenarios, scientists have developed physical C cycle models, which describe the physical inter-dependencies between C stores, plant function, soil water and energy cycling, and dead organic C storage.

There are key challenges in utilizing carbon cycle models to advance and retrieve quantitative knowledge about C cycling in the Earth's biosphere. A principal challenge lies in the inherent complexity of the terrestrial ecosystems and its subsequent representation in carbon cycle models. C dynamics are governed by highly nonlinear processes and emergent environmental sensitivities that vary widely over space and time. These dynamics are attributable to a wealth and complexity of biological function across ecosystem processes (such as leaf area responses to climate, plant C allocation, disturbance-recovery dynamics and soil microbial decomposition processes) which are tightly interconnected yet are challenging to model and validate.

2.2 Carbon Cycle Science Visual Analytics

Researchers investigating visual analysis of the C cycle generally have elaborated on combinations of three different modes of visual representation. First, researchers represent a single state of the model. Spatial multi-variate data are often viewed as map overlays [17], or shown as linear, bar or polar charts [26], organized into dashboard-like small multiple displays [10]. To focus on key variables, some C models [5] compose scenes that map a reduced number of dimensions onto iconified glyphs, creating a kind of 2D cinematic information graphic. Second, researchers consider relationships between elements of the model. Graph formalization of relationships appears broadly in the sciences, such as capacitance, water, and biology more broadly [9, 25]. Such formalizations appear in C cycle research to represent individual zone transfers, statistics, and probes on specific points of interest in models [24]. Third, researchers represent states of a model as a series over multiple time steps. In spatial data, this comparison has been explored as combinations of small multiples of single views [23], or transitions using

time sliders [7], or animation [2]. To facilitate exploration and discovery, CLOVE juxtaposes all three modes of representations into a single visual analytic application.

3 DESIGN STUDY

This paper shares a design study [21] of C cycle scientists, and the CLOVE application designed to support their analysis of the C cycle with the CARDAMOM framework. As part of the Data to Discovery program [13], the design team spent 10 weeks, approximately 40 hours each week, engaged with the scientists. We began with contextual inquiry [4], observing work processes in context in two-hour sessions, identifying potential friction points, and developing an understanding of the users' data and tasks. The design team then transitioned to lightweight prototyping [1], in weekly 90-minute meetings with the scientists to evaluate and co-design prototypes [20]. Sketches were first created on paper in individual and group ideation sessions, then shared with mentors via in-person pin-ups and Miro boards for critique. These sessions informed higher-fidelity prototypes built with tools including p5.js and TouchDesigner. Design artifacts - including sketches, code iterations, and feedback notes - were archived in a shared digital repository to document evolving rationale. The final phase of the design study included weekly think-aloud evaluations of more mature prototypes with the users [18]. The software prototype described later in this paper was iteratively developed in Javascript using D3 [6].

3.1 User Tasks

Scientists using the CARDAMOM framework aim to develop accurate, observation-informed representations of Earth's C, water, and energy cycles in order to address critical questions in carbon cycle science and its role in the Earth system. We organize these goals into a number of analytic tasks that scientists perform:

Task 1. Analyze how C moves through ecosystems over time by tracking storage, transfer, and release across pools (e.g., foliage, wood, soil).

Task 2. Identify drivers of C flux changes, distinguishing natural cycles (respiration, decomposition) from disturbances (wildfires, land-use change).

Task 3. Measure C residence times to evaluate sequestration efficiency and determine if ecosystems act as C sources or sinks.

Task 4. Compare seasonal cycles and anomalies (droughts, fires) to improve predictive model accuracy.

Task 5. Detect long-term shifts in C dynamics by examining multi-year patterns in storage and emissions.

3.2 CARDAMOM Variables and Data Structures

The dataset used in this study comprises outputs from the CARDAMOM framework (see [5] for a full data description), offering monthly observations of C dynamics at a specific location from January 2001 to December 2021. The selected variables were identified through discussion with CARDAMOM researchers, and transformed into tabular format for downstream analysis. At each time step, the full model includes more than 100 continuous variables representing model states across categories that include: (1) **C pools** such as foliar C, wood C, root C, labile C, litter C, and soil organic C; (2) **fluxes** such as gross primary production, autotrophic respiration, heterotrophic respiration, net primary production, litter-fall and soil respiration; (3) **disturbance events** that include carbon loss due to fire and ecosystem recovery following combustion; and (4) **water and energy fluxes** that include evapotranspiration, soil moisture and heat fluxes. At each time step the data are arranged in a network that includes continuous variables that represent directed C flows across and in-between states.

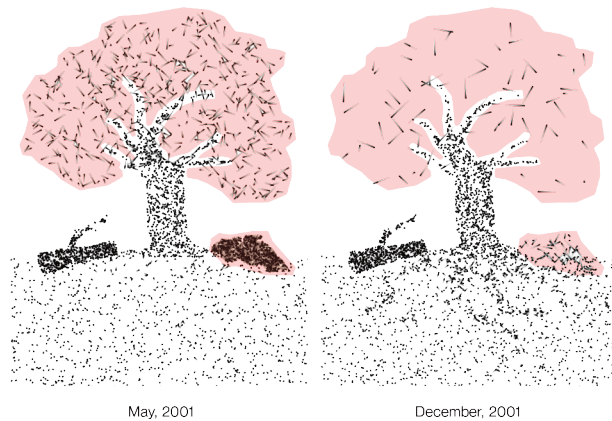


Figure 2: Two views from CLOVE's Residence Time Tree view, representing carbon dynamics in May and December 2001. The pink-highlighted regions correspond to different substance pools of the carbon cycle. Particle density reflects the quantity of carbon, while particle speed represents the turnover rate for each pool.

4 THE CLOVE APPLICATION

The CLOVE application presents a series of innovative interactive data visualizations designed specifically to facilitate the exploration and interpretation of CARDAMOM's complex, high-dimensional C cycle outputs. To accomplish this task, CLOVE provides a set of linked views designed to support key C cycle science tasks. In this section, we describe each of those views, and how they support the tasks specified in Section 3.1.

4.1 Residence Time Tree

The Residence Time Tree (Figure 2) introduces a novel visual metaphor to communicate C storage and turnover within terrestrial ecosystems (Tasks 1, 2). In this view, C pools - such as foliage, wood, roots, litter, and soil - are spatially organized into a scene around a small, hierarchically connected ecosystem, situated as a literal tree, reflecting the underlying spatial and ecological relationships between C pools. C quantities within each pool are represented dynamically through particle counts, while the turnover rates, the speed at which C moves in and out, are conveyed by particle speed. Faster moving particles signify shorter residence times and higher turnover rates, enabling users to quickly grasp otherwise abstract C dynamics through direct visual perception.

This view is designed to anchor complex ecological data within a familiar visual schema, facilitating rapid comprehension and recall. For example, in May 2001 (Figure 3, left), there is a high concentration of active particles in the leafy foliage and litter pools, whereas in December of 2021 (Figure 3, right), activity shifts toward the fine root area. Within the application, such comparisons can be modified to encode seasonal, monthly, or annual patterns. By integrating both particle quantity and velocity, the visualization is designed to convey a more layered and dynamic narrative than static images, potentially enabling more nuanced interpretations of C flow.

4.2 Shell Relationship Diagram

The Shell Relationship Diagram (Figure 3) is a directional chord diagram (or Kruskogram [27]) designed to reveal how C moves between ecosystem components. C pools appear as vertically stacked circular nodes, while C fluxes are visualized as curved chords. Flux direction is spatially encoded. Left-side chords represent C moving into pools (e.g., from the atmosphere into foliage via photosynthesis). Right-side chords represent C moving out of pools (e.g.,

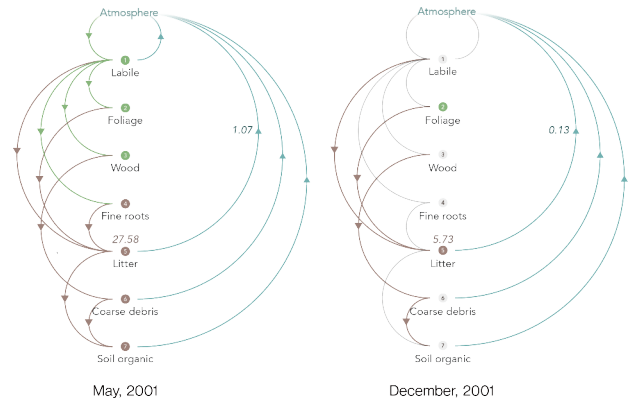


Figure 3: Two views from CLOVE's Shell Relationship Diagram, also showing May (left) and December (right) 2001. This more abstract, hierarchical overview focuses attention on the model's aggregate architecture, highlighting not just present flows but absent ones. May shows higher foliage than December, similar to the Residence Time Tree (Figure 2).

respiration returning C to the atmosphere).

This view supports Tasks 1, 2, and 4 by enabling users to trace both the magnitude and directionality of C transfers over time. For example, in May 2001 (Figure 3, left), downstream and upstream flows are visible across most pools, signaling active processes like photosynthesis, decomposition, and respiration. By December of 2001 (Figure 3, right), several chords fade to gray, indicating that certain C exchanges have slowed or ceased altogether. This seasonal shift is particularly evident in the foliage pool, which shows outgoing brown arrows in December but no incoming green arrows - suggesting C is leaving without replenishment. This pattern aligns with the drop in particle density seen in the Residence Time Tree (Figure 2). The Shell Diagram offers a dynamic view of carbon flow, emphasizing when and where carbon moves between pools. By choosing not to overload the arc or node thickness with scalar values of C flux or store, the diagram serves instead to highlight the presence and directionality of the flux. This view also highlights different patterns than the Residence Time Tree, which instead shows how much carbon remains in each pool.

4.3 Medal Relationship Diagram

The Medal Relationship Diagram (Figure 4) offers a visualization of relationships among C pools and fluxes. It supports Tasks 1, 2, and 4 by enabling spatial reasoning across C categories and highlighting temporal anomalies in flux behavior. While conceptually aligned with the Shell Diagram, the Medal Diagram introduces explicit spatial grouping - organizing C pools into functional regions such as the atmosphere, live biomass, and soil, based on their physical or ecological proximity.

4.4 Aggregate Flux Petal Plot

The Aggregate Flux Petal Plot (Figure 5) visualizes changes in C flux categories - such as respiration, production, decomposition, fire, and disturbance - through intuitive visual metaphors. This visualization aids Tasks 2, 4, and 5 by helping researchers compare current flux values to historical norms and detect abnormal seasonal behaviors. Each flux category is represented by an individual petal, encoded with distinct visual characteristics to communicate its value at specific timestamps. Colored-filled petals without stroke outlines indicate the current flux category values, while dotted-stroke petals represent monthly averages, and solid-stroke

petals depict overall averages. This encoding allows researchers to quickly compare fluxes to historical norms, enabling the rapid identification of deviations and seasonal patterns within the C cycle.

4.5 Garden Petal Plot

Expanding on the petal plot concept, the Garden Petal Plot (Figure 6) uses a small multiples layout to show how C fluxes evolve over time. Each miniature petal plot represents a monthly snapshot, with petal size indicating the magnitude of each C flux category.

By arranging these petal plots chronologically across months and years, the diagram supports Tasks 4 and 5 by enabling pattern recognition, seasonal comparison, and anomaly detection. The repeated structure resembles a “garden” of C behavior, allowing researchers to track fluctuations across respiration, production, disturbance, fire, and decomposition in a highly compact form.

For example, the highlighted plot in July 2002 shows a prominent red petal, indicating a spike in fire-related fluxes. Additionally, purple disturbance petals are visible from April through July of that year, whereas 2001 and 2003 show minimal or no disturbance activity. This visual pattern reveals heightened disturbance and fire events during the 2002 growing season, thereby enabling isolation and future investigation of this usual period within the broader C cycle.

5 USE CASE AND EVALUATION

Figure 1 describes an analysis that we draw from our observations with scientists, and features the application, loaded with data from a major fire-related event in June, 2002. Using the interface, our scientist uses the residence time tree (left, bottom) to understand the storage and fluctuation of carbon within multiple elements of the environment. They then use the aggregated flux petal plot (center) to break down the carbon system, and find an abnormal rate for fire, which they can contextualize using the garden petal plot (bottom). Based on feedback from CARDAMOM scientists, CLOVE’s interactive visualizations helped them understand data in new ways. For example, one researcher described how the residence time tree, which animates carbon turnover rates across different pools, enabled them to perceive dynamic rates of change that would otherwise require manual calculations and side-by-side number comparisons over time. Finally, they use the Shell (bottom, right) and Medal (top, right) Relationship Diagrams to see which

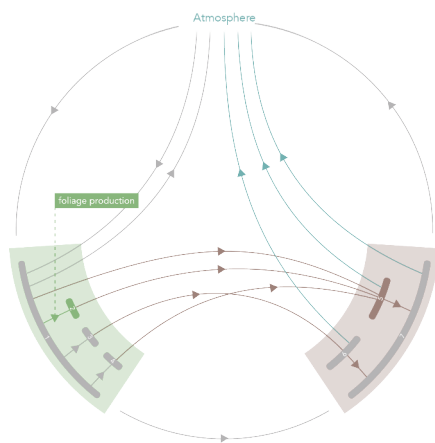


Figure 4: CLOVE’s Medal relationship diagram provides a higher-level overview than the Shell Diagram (Figure 3), introducing a hierarchical grouping of organic pools while preserving directional carbon flow between them.

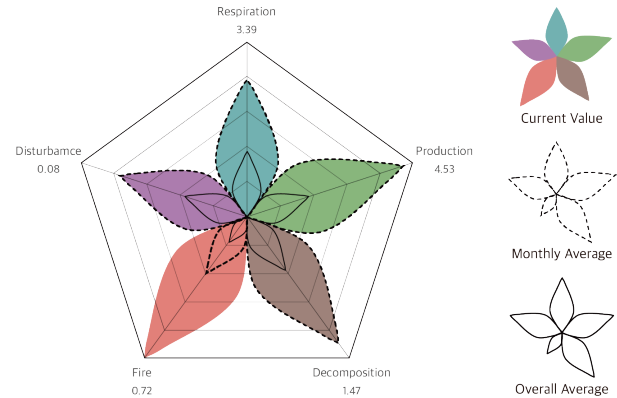


Figure 5: The Aggregate Flux Petal Plot compares current carbon flux values (filled petals) with historical monthly averages (dashed lines) and overall averages (solid lines) across five categories: respiration, production, decomposition, fire, and disturbance.

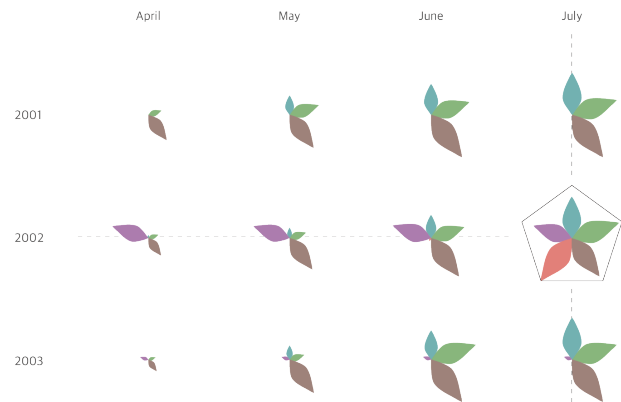


Figure 6: Garden Petal Plot presenting a chronological grid of monthly carbon flux snapshots across multiple years. Each petal represents a distinct flux category, with petal size indicating magnitude. The layout enables quick visual comparison across time to reveal seasonal trends, year-to-year variation, and unusual flux behavior.

carbon sources are being impacted by fire, releasing carbon into the atmosphere.

6 CONCLUSION

Advancing C cycle science requires interpreting high-dimensional datasets capturing complex C flux, storage, and turnover interactions. This design study presents CLOVE, a visual analytics system to support such inquiry. We explore how its Residence Time Tree, Shell Relationship Diagram, Medal Relationship Diagram, Aggregate Flux Petal Plot, and Garden Petal Plot enable users to identify patterns, track flux, detect seasonal cycles, and uncover long-term trends. This interactive approach lowers barriers to understanding, facilitating exploration and hypothesis generation for both model users and developers.

Moving forward, we see the opportunity to explore how CLOVE could incorporate a wide range of land models with compliant C cycle dynamics (e.g. the TRENDY model ensemble [22]), and add capabilities for uncertainty visualization and collaborative exploration to support broader applications in Earth system modeling. We anticipate that these tools will serve scientists to both better constrain current dynamics of ecosystem C cycling, and to ultimately better understand terrestrial C exchanges in the past, present, and future.

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